

DEVOPS PROGRAM 1

Title: Build a Docker Container from a Custom Dockerfile

Objective:

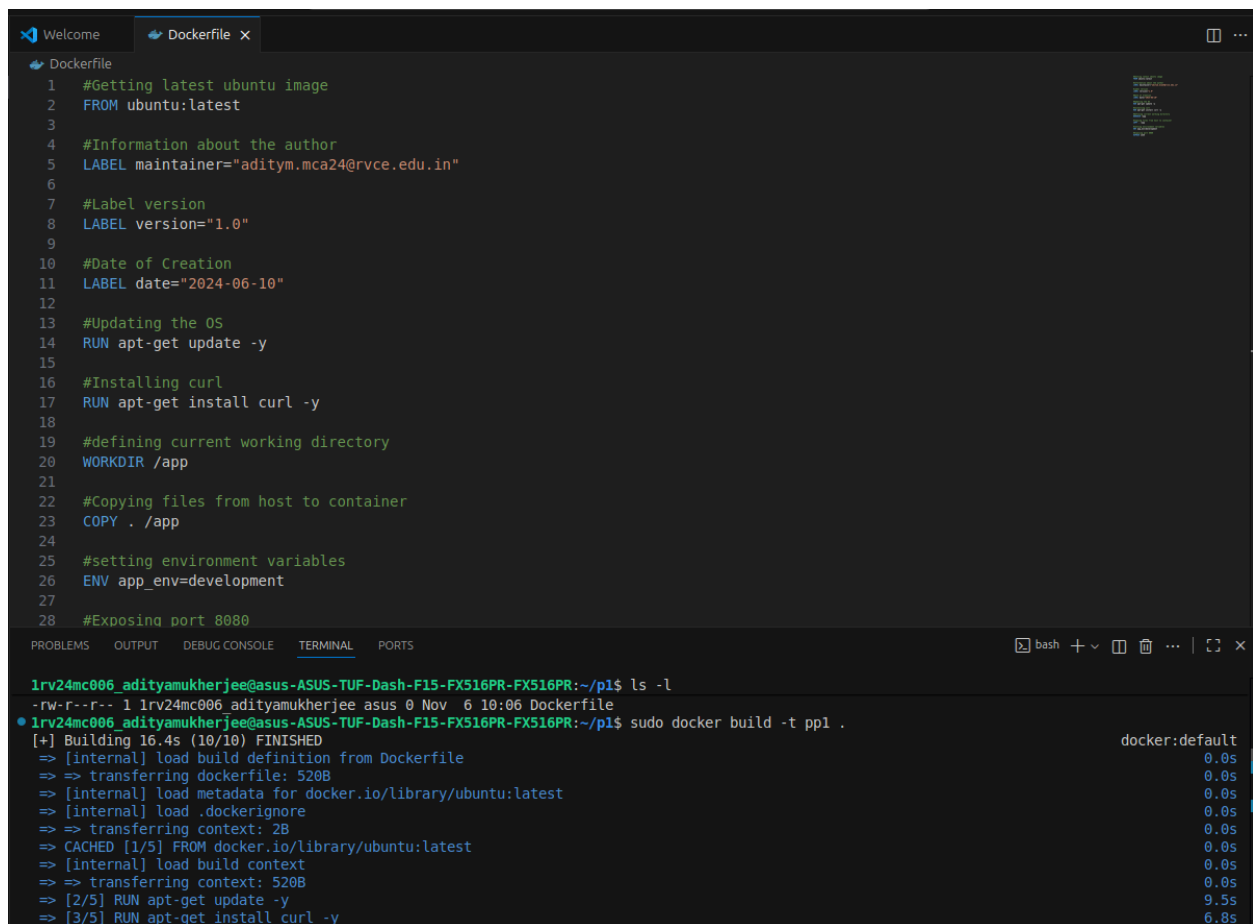
To create a custom Docker image from a user-provided Dockerfile (based on Ubuntu) and run it inside a Docker container, verifying the build and run process

Software Requirements:

- Docker
- VS code(for editing)

Files Used:

1. Dockerfile – Custom Dockerfile based on ubuntu:latest



The screenshot shows a Visual Studio Code editor with a Dockerfile open. The Dockerfile contains instructions to build a container based on Ubuntu, including setting labels, updating the OS, installing curl, and setting environment variables. Below the editor, the terminal shows the command `sudo docker build -t pp1 .` being executed, followed by the build progress and completion status.

```
Dockerfile
1 #Getting latest ubuntu image
2 FROM ubuntu:latest
3
4 #Information about the author
5 LABEL maintainer="aditym.mca24@rvce.edu.in"
6
7 #Label version
8 LABEL version="1.0"
9
10 #Date of Creation
11 LABEL date="2024-06-10"
12
13 #Updating the OS
14 RUN apt-get update -y
15
16 #Installing curl
17 RUN apt-get install curl -y
18
19 #defining current working directory
20 WORKDIR /app
21
22 #Copying files from host to container
23 COPY . /app
24
25 #setting environment variables
26 ENV app_env=development
27
28 #Exposing port 8080

1rv24mc006_adityamukherjee@asus-ASUS-TUF-Dash-F15-FX516PR-FX516PR:~/pp1$ ls -l
-rw-r--r-- 1 1rv24mc006_adityamukherjee asus 0 Nov  6 10:06 Dockerfile
1rv24mc006_adityamukherjee@asus-ASUS-TUF-Dash-F15-FX516PR-FX516PR:~/pp1$ sudo docker build -t pp1 .
[+] Building 16.4s (10/10) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 520B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [1/5] FROM docker.io/library/ubuntu:latest 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 520B 0.0s
=> [2/5] RUN apt-get update -y 9.5s
=> [3/5] RUN apt-get install curl -y 6.8s
docker:default
```

Procedure:

1. Create project folder and place Dockerfile and any app files inside it.
2. Open a terminal in the project folder.

3. Build the Docker image:
 `sudo docker build -t pp1 .`
4. Run the Docker container:
 `sudo docker run -d -p 8080:8080 pp1`
5. Verify running containers:
 `sudo docker ps`
6. Access application at:
 <http://localhost:8080>

Dockerfile Summary:

```
FROM ubuntu:latest
LABEL maintainer="aditym.mca24@rvce.edu.in"
LABEL version="1.0"
LABEL date="2024-06-10"
RUN apt-get update -y
RUN apt-get install curl -y
WORKDIR /app
COPY . /app
ENV app_env=development
EXPOSE 8080
```

Result:

A Docker image named 'pp1' was built successfully and a container was started with port 8080

Conclusion:

This exercise demonstrates how to create a Docker image from a custom Dockerfile and run it as a container.